

PY1010 — Physics for Biological and Chemical Sciences

Summer 2022 Examination · Paper 1 · Full Marking Scheme

Module	PY1010 — Physics for Biological and Chemical Sciences
Internal examiners	Dr John Cuffe, Prof. Paul Callanan
Institution	University College Cork
Session	Summer 2022, Paper 1
Duration	3 hours (+ 15 minutes reading time)
Structure	Six questions, each given as an either/or choice (A or B). Answer one option of each of the six questions and all parts of that option. Marks for each part are shown in square brackets.
Materials	Non-programmable calculator permitted; constants on the paper's "useful data" page

How to use this scheme. Every answer is worked out in small numbered steps, with a "Rules you need" box listing the formulas used. Nothing is skipped, so you can see where every number comes from: every unit conversion, every substitution, every check. Read the answer first, then follow the steps.

Coverage note. This scheme works through **both options (A and B) of all six questions** in full. On the day you only choose one option per question, but working both lets you pick the one you're strongest on.

Source basis. Every answer is built **only** from the PY1010 lecture material given to students: the lecture slides (mechanics, fluids, thermodynamics) and the handwritten lecture notes (gas laws, electricity, magnetism, waves and optics). No textbook or outside material is used, except where a highlighted note says the lecture notes don't cover something (Q5(B)(c) and Q6(A)(a)).

Own-figure rule. Where a part carries forward a number from an earlier part, an answer that uses the correct method on an earlier wrong value still earns the method marks for the later part.

Numbers. $g = 9.81 \text{ m s}^{-2}$ and other constants from the paper's useful-data page. Intermediate values are kept unrounded; final answers are given to 2–3 significant figures. Answers differing only through rounding get full marks.

Question 1 · Option A

Order-of-magnitude estimates: grains of sand vs stars in the universe — 10 marks

Given: 30% of the world's coastline is sandy; ~1 million km of coastline; beaches 50 m wide and 25 m deep; each grain of sand occupies 1 mm^3 .

1(A)(a)

[1 mark]

Estimate the total length of sandy beaches on Earth, in km.

ANSWER

$\approx 3 \times 10^5 \text{ km of sandy beach}$

RULES YOU NEED

- 30% of something = $0.30 \times \text{it}$

WORKING — STEP BY STEP

1 Take 30% of the coastline

$$L = 0.30 \times 1\,000\,000 \text{ km} = 300\,000 \text{ km} = 3 \times 10^5 \text{ km}$$

Marks	Awarded for
1	$3 \times 10^5 \text{ km.}$

1(A)(b)**[3 marks]**

If the beaches span 50 m from land to sea and are 25 m deep, what is the total volume of sand, in m^3 ?

ANSWER

$$V \approx 3.8 \times 10^{11} \text{ m}^3$$

RULES YOU NEED

- Volume of a box = length $\times$ width $\times$ depth
- 1 km = 10^3 m

WORKING — STEP BY STEP**1 Treat all the beaches as one long box**

Length = the beach length from (a), width = 50 m, depth = 25 m.

2 Convert the length to metres

$$3 \times 10^5 \text{ km} = 3 \times 10^5 \times 10^3 \text{ m} = 3 \times 10^8 \text{ m}$$

3 Multiply the three sides

$$V = 3 \times 10^8 \times 50 \times 25 = 3 \times 10^8 \times 1250 = 3.75 \times 10^{11} \text{ m}^3$$

Marks	Awarded for
1	Length converted to metres.
1	$V = \text{length} \times \text{width} \times \text{depth}$.
1	$\approx 3.8 \times 10^{11} \text{ m}^3$.

1(A)(c)**[3 marks]**

Hence, estimate how many grains of sand there are in all the beaches in the world.

ANSWER

$N \approx 4 \times 10^{20}$ grains of sand

RULES YOU NEED

- $1 \text{ mm} = 10^{-3} \text{ m}$, so $1 \text{ mm}^3 = 10^{-9} \text{ m}^3$
- Number = total volume $\div$ volume of one grain

WORKING — STEP BY STEP**1 Volume of one grain in m^3**

$$1 \text{ mm}^3 = (10^{-3} \text{ m})^3 = 10^{-9} \text{ m}^3$$

2 Divide

$$N = 3.75 \times 10^{11} \div 10^{-9}$$

3 Dividing by 10^{-9} = multiplying by 10^9

$$N = 3.75 \times 10^{11+9} = 3.75 \times 10^{20} \text{ grains}$$

EXAM TECHNIQUE

Cube the conversion factor: $1 \text{ mm}^3 = 10^{-9} \text{ m}^3$, not 10^{-3} m^3 .

Marks	Awarded for
1	$1 \text{ mm}^3 = 10^{-9} \text{ m}^3$.
1	$N = V_{\text{total}}/V_{\text{grain}}$.
1	$\approx 4 \times 10^{20}$.

1(A)(d)**[3 marks]**

There are $\sim 10^{11}$ galaxies, each with $\sim 10^{11}$ stars. How many stars are there in total, and how does it compare to the grains of sand?

ANSWER

10^{22} stars: about 25-30 times more stars than grains of sand

RULES YOU NEED

- $10^a \times 10^b = 10^{a+b}$

WORKING — STEP BY STEP**1 Total number of stars**

$$N_{\text{stars}} = 10^{11} \times 10^{11} = 10^{22}$$

2 Compare with the sand

$$N_{\text{stars}} / N_{\text{sand}} = 10^{22} \div 3.75 \times 10^{20} = 26.7$$

3 Conclusion

There are roughly **25-30 times more stars** in the universe than grains of sand on all the world's beaches.

Marks	Awarded for
1	10^{22} stars.
1	Ratio calculated.
1	Conclusion: more stars than grains of sand (own-figure from (c) accepted).

Question 1(A) Total: 10 Marks

Question 1 · Option B

Speed vs velocity, kinematic equations, road bowling — 10 marks

Given: Road-bowling ball travels 60 m in 5 s, decelerating at a constant rate to rest;
Viaduct Bridge height 28 m.

1(B)(a)

[2 marks]

Define and give the SI units for (i) speed (ii) velocity.

ANSWER

Speed = distance/time (scalar); velocity = displacement/time (vector); both in m s^{-1}

RULES YOU NEED

- Speed = distance ÷ time
- Velocity = displacement ÷ time (a vector)

WORKING — STEP BY STEP

1 Speed

Total distance ÷ total time. It has a size only (a scalar). Unit: m s^{-1} .

2 Velocity

Displacement ÷ time, $v = \Delta x / \Delta t$. It has a size **and a direction** (a vector). Unit: m s^{-1} .

Marks	Awarded for
1	Speed defined, m s^{-1} .
1	Velocity as a vector (displacement/time), m s^{-1} .

1(B)(b)**[1 mark]**

Under what condition are the Kinematic Equations of Motion valid?

ANSWER

Only when the acceleration is constant

RULES YOU NEED

• $v = v_0 + at$; $x = x_0 + v_0t + \frac{1}{2}at^2$; $v^2 = v_0^2 + 2a(x - x_0)$

WORKING — STEP BY STEP**1 The condition**

They only hold when the acceleration is **constant** ($a = \text{constant}$).

Marks**Awarded for****1**

Constant acceleration.

1(B)(c)(i)**[3 marks]**

A ball travels 60 m in 5 s before rolling to a halt, decelerating at a constant rate. Calculate its initial velocity.

ANSWER

$$v_0 = 24 \text{ m s}^{-1}$$

RULES YOU NEED

- Constant acceleration: $x = \frac{1}{2} (v + v_0) t$

WORKING — STEP BY STEP**1 List what you know**

Distance $x = 60 \text{ m}$, time $t = 5 \text{ s}$, final speed $v = 0$ (it stops). We want v_0 .

2 Pick the equation with x , t , v and v_0

$$x = \frac{1}{2} (v + v_0) t$$

3 Substitute

$$60 = \frac{1}{2} \times (0 + v_0) \times 5 = 2.5 v_0$$

4 Solve

$$v_0 = 60 \div 2.5 = 24 \text{ m/s}$$

Marks	Awarded for
1	$v = 0$ at the end; suitable equation chosen.
1	Correct substitution.
1	24 m s^{-1} .

1(B)(c)(ii)**[4 marks]**

He throws the ball vertically upwards at the velocity from (i), next to a 28 m bridge. What maximum height will it reach? Could it pass over the bridge?

ANSWER

$h \approx 29.4$ m: just possible, but only by ≈ 1.4 m

RULES YOU NEED

- Going up: $a = -g$
- At the top: $v = 0$
- $v^2 = v_0^2 + 2a(x - x_0)$

WORKING — STEP BY STEP**1 List what you know**

$v_0 = 24$ m/s upwards, $a = -9.81$ m/s² (gravity slows it), and at the highest point $v = 0$.

2 Use $v^2 = v_0^2 + 2ah$

$$0 = 24^2 - 2 \times 9.81 \times h$$

3 Solve for h

$$h = 24^2 \div (2 \times 9.81) = 576 \div 19.62 = 29.4 \text{ m}$$

4 Compare with the bridge

$29.4 \text{ m} > 28 \text{ m}$, so in principle it just clears the bridge, by about 1.4 m. Air resistance (ignored here) would lower it, so in practice it would be very close.

Marks	Awarded for
1	$a = -g$ and $v = 0$ at the top.
1	$v^2 = v_0^2 + 2ah$ rearranged.
1	$h \approx 29.4$ m (own-figure accepted).
1	Compares with 28 m and concludes (just) possible.

Question 1(B) Total: 10 Marks

Question 2 · Option A

Power, energy cost of accelerating a car, driving speed and fuel use — 10 marks

Given: Car of mass 1000 kg, $0 \rightarrow 100$ km/hr in 10 s. Petrol €2.00 per litre, 10^7 J per litre.
Electricity 30 c per kW hr.

2(A)(a)

[2 marks]

Define power, and give its SI unit.

ANSWER

$P = W/t$; unit the watt, $W = J\ s^{-1}$

RULES YOU NEED

- $P = W / t$

WORKING — STEP BY STEP

1 Definition

Power is the rate of doing work (or of using energy): $P = W/t$.

2 Unit

Joules per second, $J\ s^{-1}$, called the **watt (W)**.

Marks	Awarded for
1	Rate of doing work, $P = W/t$.
1	Watt ($J\ s^{-1}$).

2(A)(b)(i)**[2 marks]**

If petrol costs €2.00 per litre, and each litre gives 10^7 J to the engine, how much does it cost to accelerate the car to 100 km/hr (in cent)?

ANSWER

≈ 7.7 cent (assuming all the fuel energy becomes kinetic energy)

RULES YOU NEED

- Work done = gain in kinetic energy = $\frac{1}{2}mv^2$
- 100 km/hr $\div 3.6 =$ m/s

WORKING — STEP BY STEP**1 Speed in m/s**

$$v = 100\,000 \text{ m} \div 3600 \text{ s} = 27.78 \text{ m/s}$$

2 Energy needed = kinetic energy gained

$$E = \frac{1}{2} \times 1000 \times 27.78^2 = \frac{1}{2} \times 1000 \times 771.6 = 3.86 \times 10^5 \text{ J}$$

3 Litres of petrol

$$3.86 \times 10^5 \text{ J} \div 10^7 \text{ J per litre} = 0.0386 \text{ L}$$

4 Cost

$$0.0386 \text{ L} \times 200 \text{ c per litre} = 7.7 \text{ c}$$

(This assumes all the petrol's energy goes into kinetic energy.)

Marks	Awarded for
1	$E_{\text{kin}} = \frac{1}{2}mv^2$ with v in m s^{-1} .
1	≈ 7.7 c.

2(A)(b)(ii)**[2 marks]**

If the engine were electric, how much would that cost instead, at 30 c per kW hr?

ANSWER

≈ 3.2 cent

RULES YOU NEED

- Energy = power × time
- 1 kW hr = 1000 W × 3600 s = 3.6×10^6 J

WORKING — STEP BY STEP**1 How many joules is a kW hr?**

$$1 \text{ kW hr} = 1000 \text{ J/s} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}$$

2 Energy in kW hr

$$3.86 \times 10^5 \text{ J} \div 3.6 \times 10^6 \text{ J} = 0.107 \text{ kW hr}$$

3 Cost

$$0.107 \times 30 \text{ c} = 3.2 \text{ c}$$

Marks	Awarded for
1	1 kW hr = 3.6×10^6 J.
1	≈ 3.2 c.

2(A)(c)(i)**[2 marks]**

Cars A (80 km/hr) and B (100 km/hr) each travel 100 km. If power used $\propto v^2$, how much more power does B use than A?

ANSWER

About 1.56 times the power ($\approx$ 56% more)

RULES YOU NEED

- $P \propto v^2$, so $P_B/P_A = (v_B/v_A)^2$

WORKING — STEP BY STEP**1 Ratio of speeds**

$$v_B / v_A = 100 / 80 = 1.25$$

2 Square it

$$P_B / P_A = 1.25^2 = 1.5625$$

3 Interpret

Car B uses about 1.56 times the power: **56% more**.

Marks	Awarded for
1	$P \propto v^2$, ratio $(100/80)^2$.
1	1.56 (56% more).

2(A)(c)(ii)**[2 marks]**

How much more energy does car B use, relative to car A?

ANSWER**1.25 times the energy (25% more)****RULES YOU NEED**

- Energy = power × time
- Time for a fixed distance: $t = d / v$

WORKING — STEP BY STEP**1 Times for the 100 km**

$$t_A = 100/80 = 1.25 \text{ h}; t_B = 100/100 = 1.00 \text{ h}$$

So B's time is 0.8 of A's.

2 Ratio of energies

$$E_B / E_A = (P_B / P_A) \times (t_B / t_A) = 1.5625 \times 0.8 = 1.25$$

3 InterpretB uses 1.25 times the energy: **25% more**. Driving more slowly does save fuel.

Marks	Awarded for
1	$E = Pt$ with $t = d/v$.
1	1.25 (25% more).

Question 2(A) Total: 10 Marks

Question 2 · Option B

Centripetal force and acceleration, the International Space Station — 10 marks

Given: ISS orbits 390 km above the surface; mass 4.5×10^5 kg; take g the same as at the Earth's surface; Earth radius 6400 km.

2(B)(a)

[1 mark]

Explain what is meant by centripetal force.

ANSWER

The net force, directed towards the centre, needed to keep an object moving in a circle

RULES YOU NEED

- Uniform circular motion is always accelerated (direction of v keeps changing)
- $F = ma$

WORKING — STEP BY STEP

1 Why a force is needed

In circular motion the direction of the velocity keeps changing, so the object is always accelerating towards the centre.

2 Definition

The centripetal force is the **net force directed towards the centre** of the circle that causes this acceleration and keeps the object on its circular path.

Marks

Awarded for

1

Net force towards the centre that keeps the object in circular motion.

2(B)(b)**[1 mark]**

The centripetal force is mv^2/r . Write down an expression for the acceleration due to this force.

ANSWER

$a = v^2/r$ (directed towards the centre)

RULES YOU NEED

- $F = m a$, so $a = F / m$

WORKING — STEP BY STEP**1 Divide the force by the mass**

$$a = (m v^2 / r) \div m = v^2 / r$$

Directed towards the centre.

Marks**Awarded for****1** $a = v^2/r$.

2(B)(c)(i)**[1 mark]**

What thrust is required from its engines to keep the ISS in orbit?

ANSWER

Zero: gravity alone supplies the centripetal force

RULES YOU NEED

- Newton's first law: no force is needed to keep moving at constant speed

WORKING — STEP BY STEP**1 What keeps it on the circle?**

Gravity provides the centripetal force that bends its path.

2 Is a forward push needed?

No. By Newton's first law it keeps its speed without any force along its motion.

Thrust needed: **zero** (ignoring any air drag).

Marks	Awarded for
1	Zero thrust, with reason.

2(B)(c)(ii)**[2 marks]**

What gravitational force (in N) acts on the ISS?

ANSWER

$F \approx 4.4 \times 10^6 \text{ N}$

RULES YOU NEED

• $w = m g$

WORKING — STEP BY STEP**1 Weight with g as at the surface**

$$F = m g = 4.5 \times 10^5 \times 9.81 = 4.41 \times 10^6 \text{ N}$$

Marks	Awarded for
1	$F = mg$ ($w = mg$).
1	$4.4 \times 10^6 \text{ N}$.

2(B)(c)(iii)**[2 marks]**

What is (A) the magnitude and (B) the direction of the centripetal force acting on the ISS?

ANSWER**(A) 4.4×10^6 N; (B) towards the centre of the Earth****RULES YOU NEED**

- The centripetal force is whatever force pulls the object towards the centre

WORKING — STEP BY STEP**1 Which force is it?**

Gravity is the only force, and it points to the Earth's centre, so the centripetal force **is** the gravitational force.

2 Answer

(A) 4.4×10^6 N; (B) towards the centre of the Earth.

Marks	Awarded for
1	Magnitude 4.4×10^6 N.
1	Direction: towards the Earth's centre.

2(B)(c)(iv)**[2 marks]**Calculate the speed (in m s^{-1}) at which the ISS orbits the Earth.**ANSWER**

$$v \approx 8.2 \times 10^3 \text{ m s}^{-1} \text{ (8.2 km s}^{-1}\text{)}$$

RULES YOU NEED

- Gravity = centripetal force: $mg = mv^2/r$
- r is measured from the Earth's centre

WORKING — STEP BY STEP**1 Orbit radius**

$$r = 6400 \text{ km} + 390 \text{ km} = 6790 \text{ km} = 6.79 \times 10^6 \text{ m}$$

2 Set gravity equal to the centripetal force

$$m g = m v^2 / r \Rightarrow v^2 = g r$$

3 Calculate

$$v^2 = 9.81 \times 6.79 \times 10^6 = 6.66 \times 10^7$$

$$v = \sqrt{(6.66 \times 10^7)} = 8.16 \times 10^3 \text{ m/s}$$

Marks	Awarded for
1	$r = R_E + 390 \text{ km}$ and $mg = mv^2/r$.
1	$\approx 8.2 \times 10^3 \text{ m s}^{-1}$.

2(B)(c)(v)**[1 mark]**

If the centripetal force were switched off, what would happen to the ISS?

ANSWER

It would fly off in a straight line along the tangent to its orbit

RULES YOU NEED

- Newton's first law

WORKING — STEP BY STEP**1 No net force**

It would carry on at constant velocity in a **straight line**, along the tangent to its orbit at that instant, and fly off into space.

Marks**Awarded for****1**

Straight line / tangent (Newton's first law).

Question 2(B) Total: 10 Marks

Question 3 · Option A

Archimedes' principle, a helium weather balloon — 10 marks

Given: Spherical balloon, radius 2 m; $\rho_{\text{He}} = 0.18 \text{ kg m}^{-3}$; $\rho_{\text{air}} = 1.2 \text{ kg m}^{-3}$; other components' mass negligible.

3(A)(a)

[2 marks]

State Archimedes' Principle.

ANSWER

Buoyant force = weight of fluid displaced

RULES YOU NEED

- $F_{\text{buoy}} = \text{weight of fluid displaced}$

WORKING — STEP BY STEP

1 The statement

When a body is completely or partially immersed in a fluid, the fluid exerts an **upward force** equal to the **weight of the fluid displaced**. It holds for gases (like air) too.

Marks	Awarded for
1	Upward force on an immersed body.
1	Equal to the weight of fluid displaced.

3(A)(b)(i)**[2 marks]**

What upward, buoyant force does the balloon experience?

ANSWER

$$F_{\text{buoy}} \approx 394 \text{ N (upwards)}$$

RULES YOU NEED

- Sphere: $V = (4/3)\pi r^3$
- $F_{\text{buoy}} = \rho_{\text{air}} V g$

WORKING — STEP BY STEP**1 Volume of the balloon**

$$V = (4/3) \pi (2)^3 = (4/3) \pi \times 8 = 33.5 \text{ m}^3$$

2 It displaces that much air

$$F_{\text{buoy}} = 1.2 \times 33.5 \times 9.81 = 394 \text{ N}$$

Marks	Awarded for
1	$V = (4/3)\pi r^3$ and $F = \rho_{\text{air}} V g$.
1	$\approx 394 \text{ N}$.

3(A)(b)(ii)**[3 marks]**

What will the initial acceleration of the balloon be? Give magnitude and direction.

ANSWER

$a \approx 56 \text{ m s}^{-2}$, vertically upwards

RULES YOU NEED

- Net force = buoyancy – weight
- $F = m a$

WORKING — STEP BY STEP**1 Mass of the helium (the only mass)**

$$m = \rho_{\text{He}} V = 0.18 \times 33.5 = 6.03 \text{ kg}$$

2 Its weight

$$W = m g = 6.03 \times 9.81 = 59.2 \text{ N}$$

3 Net upward force

$$F_{\text{net}} = 394 - 59.2 = 335 \text{ N}$$

4 Acceleration

$$a = F_{\text{net}} / m = 335 \div 6.03 = 55.6 \text{ m/s}^2$$

Upwards. (Big, because helium is so light; in reality air resistance quickly limits the speed.)

Marks	Awarded for
1	Weight of the helium (the only mass).
1	Net force = buoyant force – weight, $a = F/m$.
1	$\approx 56 \text{ m s}^{-2}$ upwards.

3(A)(b)(iii)**[3 marks]**

As the balloon rises, the air pressure decreases. What effect does this have on the balloon's volume and the density of the gas inside?

ANSWER

The volume increases (Boyle's law, $P \propto 1/V$) and the density of the helium decreases (same mass, larger volume)

RULES YOU NEED

- Boyle's law (fixed gas, constant T): $PV = \text{constant}$
- $\rho = m / V$

WORKING — STEP BY STEP**1 Pressure inside matches outside**

As the outside pressure drops, so does the pressure of the helium.

2 Apply Boyle's law

The amount of gas is fixed and T is roughly constant, so $PV = \text{constant}$. P goes down, so **V goes up**: the balloon expands (like the rising bubble in the lectures).

3 Effect on density

Same mass of helium, bigger volume, so $\rho = m/V$ **decreases**.

Marks	Awarded for
1	Volume increases.
1	Explained via Boyle's law ($PV = \text{constant}$).
1	Density decreases ($\rho = m/V$, m fixed).

Question 3(A) Total: 10 Marks

Question 3 · Option B

Pascal's principle, pressure at depth, a hydraulic lift — 10 marks

Given: Diver at 50 m depth; clown (800 N) on a 2 cm² piston, elephant on a 60 cm² piston.

3(B)(a)

[2 marks]

State Pascal's Principle, and give the SI unit for pressure.

ANSWER

Pressure is transmitted undiminished throughout an enclosed fluid; unit the pascal, Pa = N m⁻²

RULES YOU NEED

- Pressure = force ÷ area

WORKING — STEP BY STEP

1 The statement

Pressure applied at one point in an enclosed fluid is transmitted **undiminished** to every part of the fluid and to the container walls.

2 Unit

N m⁻², called the **pascal (Pa)**.

Marks	Awarded for
1	Pascal's principle.
1	Pa (N m ⁻²).

3(B)(b)(i)**[1 mark]**

Does the diver experience pressure (a) mostly from above, (b) mostly from below, or (c) equally in all directions?

ANSWER**(c) Equally in all directions****RULES YOU NEED**

- At a given depth, fluid pressure is the same in all directions

WORKING — STEP BY STEP**1 Apply the rule**

At 50 m depth the water pushes equally from every direction.

Marks	Awarded for
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1	(c).
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3(B)(b)(ii)**[3 marks]**

Air must be supplied to the diver at ambient pressure. Calculate this pressure.

ANSWER

$$P \approx 5.9 \times 10^5 \text{ Pa (about 6 atmospheres)}$$

RULES YOU NEED

- $P = P_{\text{atm}} + \rho g h$
- $P_{\text{atm}} = 1.013 \times 10^5 \text{ Pa}$

WORKING — STEP BY STEP**1 Pressure of the water above the diver**

$$\rho g h = 1000 \times 9.81 \times 50 = 490\,500 \text{ Pa} = 4.905 \times 10^5 \text{ Pa}$$

2 Add the air pressing on the surface

$$P = 1.013 \times 10^5 + 4.905 \times 10^5 = 5.92 \times 10^5 \text{ Pa}$$

3 Sense check

That is about 6 atmospheres: every 10 m of water adds roughly 1 atmosphere ✓

Marks	Awarded for
1	$P = P_{\text{atm}} + \rho g h.$
1	$\rho g h = 4.9 \times 10^5 \text{ Pa}.$
1	$\approx 5.9 \times 10^5 \text{ Pa}.$

3(B)(c)**[4 marks]**

A clown (800 N) stands on a 2 cm² piston, connected to a 60 cm² piston with an elephant on it. Use Pascal's Principle to find the maximum mass of the elephant.

ANSWER

$$m_{\text{max}} \approx 2.4 \times 10^3 \text{ kg (about 2450 kg)}$$

RULES YOU NEED

- Pascal: equal pressure on both pistons, $F_1/A_1 = F_2/A_2$
- $w = mg$

WORKING — STEP BY STEP**1 Pressures are equal**

$$F_{\text{clown}} / A_{\text{clown}} = F_{\text{elephant}} / A_{\text{elephant}}$$

2 Rearrange for the elephant's force

$$F_{\text{elephant}} = F_{\text{clown}} \times (A_{\text{elephant}} / A_{\text{clown}})$$

3 Put the numbers in

The areas appear as a ratio, so cm² can stay as cm²:

$$F_{\text{elephant}} = 800 \times (60 / 2) = 800 \times 30 = 24\,000 \text{ N}$$

4 Weight to mass

$$m = F / g = 24\,000 \div 9.81 = 2446 \text{ kg}$$

Marks	Awarded for
1	Equal pressures on the two pistons.
1	$F_{\text{elephant}} = F_{\text{clown}} \times A$ ratio.
1	24 000 N.
1	$m = F/g \approx 2450 \text{ kg}$.

Question 3(B) Total: 10 Marks

Question 4 · Option A

Coulomb's law, forces between charges, equilibrium position, field lines — 10 marks

Given: $q_1 = +10 \mu\text{C}$ and $q_2 = +90 \mu\text{C}$, fixed at the left and right ends of a 1 m stick.

4(A)(a)

[2 marks]

State Coulomb's law for the magnitude of the force F between two point charges q_1 and q_2 a distance r apart. State the SI units of all quantities. When would F be repulsive or attractive?

ANSWER

$F = q_1 q_2 / (4\pi \epsilon_0 r^2)$; repulsive for like charges, attractive for unlike

RULES YOU NEED

- $F = q_1 q_2 / (4\pi \epsilon_0 r^2)$
- $1/(4\pi \epsilon_0) = k = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
- Like charges repel; unlike charges attract

WORKING — STEP BY STEP

1 Write the law

$$F = q_1 q_2 / (4\pi \epsilon_0 r^2) = k q_1 q_2 / r^2$$

The force gets bigger with more charge, and falls off as $1/r^2$: double the distance and the force drops to a quarter.

2 Give the unit of every symbol

F in newtons (N); q_1, q_2 in coulombs (C); r in metres (m); ϵ_0 (permittivity of free space) in $\text{C}^2 \text{ J}^{-1} \text{ m}^{-1}$.

3 Repulsive or attractive?

If q_1 and q_2 have the **same sign** (both + or both –), the force is **repulsive**. If they have **opposite signs**, it is **attractive**.

4 Check

The question asks for three things: the law, the units, and when it repels/attracts. All three are covered.

Marks	Awarded for
1	Law with SI units.
1	Like repel, unlike attract.

4(A)(b)(i)**[2 marks]**

Calculate the force on q_1 due to q_2 and on q_2 due to q_1 , and give the direction of these forces.

ANSWER

8.1 N on each: q_1 is pushed left (away from q_2), q_2 is pushed right (away from q_1)

RULES YOU NEED

- $F = k q_1 q_2 / r^2$
- $1 \mu\text{C} = 10^{-6} \text{ C}$
- Newton's third law: equal and opposite forces

WORKING — STEP BY STEP**1 Convert the charges to coulombs**

$$q_1 = 10 \mu\text{C} = 10 \times 10^{-6} \text{ C} = 1 \times 10^{-5} \text{ C}$$

$$q_2 = 90 \mu\text{C} = 90 \times 10^{-6} \text{ C} = 9 \times 10^{-5} \text{ C}$$

The distance between them is the whole stick: $r = 1 \text{ m}$.

2 Multiply the charges

$$q_1 q_2 = (1 \times 10^{-5})(9 \times 10^{-5}) = 9 \times 10^{-10} \text{ C}^2$$

3 Substitute into Coulomb's law

$$F = 8.99 \times 10^9 \times 9 \times 10^{-10} \div 1^2 = 8.09 \text{ N} \approx 8.1 \text{ N}$$

4 Directions

Both charges are positive, so they **repel**. q_1 (on the left) is pushed to the **left**; q_2 (on the right) is pushed to the **right**.

By Newton's third law the force on q_1 and the force on q_2 are the same size, 8.1 N, even though q_2 is the bigger charge.

EXAM TECHNIQUE

A common slip is thinking the bigger charge feels a bigger force. It doesn't: the same $q_1 q_2$ appears in both, so the forces are equal.

Marks	Awarded for
1	$F \approx 8.1 \text{ N}$.
1	Repulsive, equal and opposite directions.

4(A)(b)(ii)**[4 marks]**

At what point along the metre stick should a third charge be placed such that it does not move?

ANSWER

0.25 m from q_1 (0.75 m from q_2)

RULES YOU NEED

- Not moving $\Rightarrow$ net force = 0
- $F = k q_1 q_3 / r^2$
- If $a^2 = b^2$ with both positive, $a = b$

WORKING — STEP BY STEP**1 Where can it balance?**

q_1 and q_2 are both positive. Only **between** them do their forces on q_3 point in opposite directions, so that's where they can cancel. Outside the stick both forces point the same way.

2 Name the distances

Put q_3 a distance x from q_1 . Then it is $(1 - x)$ from q_2 .

3 Set the two forces equal

$$k q_1 q_3 / x^2 = k q_2 q_3 / (1 - x)^2$$

k and q_3 appear on both sides and cancel, so the answer doesn't depend on the size or sign of q_3 .

4 Rearrange

$$(1 - x)^2 / x^2 = q_2 / q_1 = 90 / 10 = 9$$

5 Take the square root

$$(1 - x) / x = 3$$

(Take the positive root, since both distances are positive.)

6 Solve for x

$$1 - x = 3x \Rightarrow 1 = 4x \Rightarrow x = 0.25 \text{ m}$$

7 Check

At 0.25 m from q_1 and 0.75 m from q_2 : $q_1/x^2 = 10/0.0625 = 160$ and $q_2/(1 - x)^2 = 90/0.5625 = 160$. Equal, so the forces balance. It makes sense that q_3 sits closer to the smaller charge.

Marks	Awarded for
1	Forces on q_3 must balance, between the charges.
1	Equation set up with x and $(1 - x)$.
1	Solved: $(1 - x)/x = 3$.
1	$x = 0.25$ m from q_1 .

4(A)(c)**[2 marks]**

Assuming q_3 is a negative charge, draw a diagram showing the electric field lines between the charges and their direction.

ANSWER

Lines leave q_1 and q_2 (positive) and end on q_3 (negative)

RULES YOU NEED

- Field lines point the way a positive test charge would be pushed
- They start on + charges and end on – charges

WORKING — STEP BY STEP**1 Direction rule**

E points in the direction of the force on a **positive** test charge. So lines point **away from** positive charges and **into** negative charges.

2 Apply it here

q_1 and q_2 are positive, so lines leave them; q_3 is negative, so lines end on it. On the outer sides of q_1 and q_2 the lines point outwards.

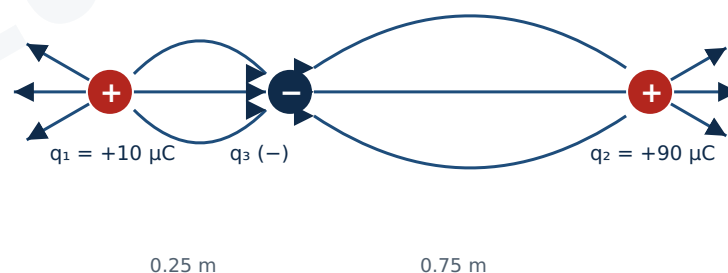


Fig. 1 — Lines leave the positive charges and end on the negative charge q_3 .

Marks	Awarded for
1	Lines from q_1 and q_2 ending on q_3 .
1	Correct arrow directions.

Question 4(A) Total: 10 Marks

Question 4 · Option B

Electric field and potential; a parallel-plate capacitor — 10 marks

Given: Plates $30\text{ cm} \times 50\text{ cm}$, separated by $d = 2\text{ mm}$, connected to a 3.7 V battery.

4(B)(a)

[2 marks]

State and explain briefly the connection between the electric field and the electric potential, and give the SI units for each.

ANSWER

$$V = Ed; E \text{ in } \text{V m}^{-1}, V \text{ in volts (V)}$$

RULES YOU NEED

- $V = W / q$ (work per unit charge)
- Uniform field: $V = E d$

WORKING — STEP BY STEP

1 What potential difference means

The potential difference V between two points is the work done per unit charge in moving a charge between them: $V = W/q$.

2 Link it to the field

In a uniform field the force on the charge is qE . Moving it a distance d along the field does work $W = qEd$. Divide by q :

$$V = E d \Leftrightarrow E = V / d$$

So the field is the potential difference per unit distance: how fast the potential changes as you move.

3 Units

E : volts per metre, V m^{-1} (the same as N C^{-1}). V : volts, $\text{V} (= \text{J C}^{-1})$.

Marks	Awarded for
1	$V = Ed$ (or $E = V/d$) with explanation.
1	Units: V m^{-1} and V .

4(B)(b)(i)**[2 marks]**

What is the magnitude of the electric field at the mid-point between the two plates?

ANSWER

$$E = 1.85 \times 10^3 \text{ V m}^{-1}$$

RULES YOU NEED

- $E = V / d$
- $1 \text{ mm} = 10^{-3} \text{ m}$

WORKING — STEP BY STEP**1 Is the mid-point special?**

No. Between parallel plates the field is **uniform**: the same everywhere between them. So the mid-point has the same field as anywhere else.

2 Convert d to metres

$$d = 2 \text{ mm} = 2 \times 10^{-3} \text{ m} = 0.002 \text{ m}$$

3 Divide

$$E = 3.7 \text{ V} \div 0.002 \text{ m} = 1850 \text{ V m}^{-1}$$

EXAM TECHNIQUE

The words “at the mid-point” are there to test you know the field is uniform. Don't halve d.

Marks	Awarded for
1	$E = V/d$ with d in m.
1	1850 V m^{-1} .

4(B)(b)(ii)**[2 marks]**

What is the total capacitance of the capacitor?

ANSWER

$$C \approx 6.6 \times 10^{-10} \text{ F (0.66 nF)}$$

RULES YOU NEED

- $C = \epsilon_0 A / d$
- $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ J}^{-1} \text{ m}^{-1}$
- $1 \text{ cm} = 0.01 \text{ m}$

WORKING — STEP BY STEP**1 Plate area in m²**

$$A = 0.30 \text{ m} \times 0.50 \text{ m} = 0.15 \text{ m}^2$$

2 Substitute

$$C = 8.854 \times 10^{-12} \times 0.15 \div 0.002$$

3 Work it out

$$8.854 \times 10^{-12} \times 0.15 = 1.328 \times 10^{-12}$$

$$1.328 \times 10^{-12} \div 0.002 = 6.64 \times 10^{-10} \text{ F}$$

EXAM TECHNIQUEConvert cm to m *before* multiplying: $30 \text{ cm} \times 50 \text{ cm} = 1500 \text{ cm}^2 = 0.15 \text{ m}^2$, not 1500 m^2 .

Marks	Awarded for
1	$C = \epsilon_0 A / d$ with A in m ² .
1	$\approx 6.6 \times 10^{-10} \text{ F}$.

4(B)(b)(iii)**[2 marks]**

A charge of $+1 \mu\text{C}$ moves from the positive to the negative plate. What is the change in electric potential energy for this charge?

ANSWER

PE decreases by $3.7 \times 10^{-6} \text{ J}$

RULES YOU NEED

- $\Delta U = q V$
- A positive charge loses potential energy moving to lower potential

WORKING — STEP BY STEP**1 Values**

$q = 1 \mu\text{C} = 1 \times 10^{-6} \text{ C}$; the potential difference between the plates is $V = 3.7 \text{ V}$.

2 Size of the change

$$\Delta U = q V = 1 \times 10^{-6} \times 3.7 = 3.7 \times 10^{-6} \text{ J}$$

3 Increase or decrease?

The positive charge goes from the positive plate (high potential) to the negative plate (low potential). It moves the way the field pushes it, like a ball rolling downhill, so its potential energy **decreases**.

Marks	Awarded for
1	$\Delta U = qV = 3.7 \times 10^{-6} \text{ J}$.
1	Stated as a decrease.

4(B)(b)(iv)**[2 marks]**

If this energy is converted to kinetic energy, and the charge has a mass of 1.87 mg, what is its final velocity? (It started from rest.)

ANSWER

$$v \approx 2.0 \text{ m s}^{-1}$$

RULES YOU NEED

- Energy lost = kinetic energy gained: $\frac{1}{2}mv^2 = \Delta U$
- $1 \text{ mg} = 10^{-6} \text{ kg}$

WORKING — STEP BY STEP**1 Mass in kg**

$$m = 1.87 \text{ mg} = 1.87 \times 10^{-3} \text{ g} = 1.87 \times 10^{-6} \text{ kg}$$

2 Set up the energy equation

$$\frac{1}{2} m v^2 = 3.7 \times 10^{-6} \text{ J}$$

3 Rearrange for v

$$v = \sqrt{(2\Delta U / m)}$$

4 Substitute

$$v = \sqrt{(2 \times 3.7 \times 10^{-6} \div 1.87 \times 10^{-6})} = \sqrt{(7.4 \div 1.87)} = \sqrt{3.96} = 1.99 \text{ m s}^{-1}$$

The powers of 10 cancel, which makes the arithmetic easy.

Marks	Awarded for
1	$\frac{1}{2}mv^2 = \Delta U$ with mass in kg.
1	$\approx 2.0 \text{ m s}^{-1}$.

Question 4(B) Total: 10 Marks

Question 5 · Option A

Lorentz force on a moving charge; force between parallel wires — 10 marks

5(A)(a)

[2 marks]

Write down an expression for F , the Lorentz force on a charge q moving at speed v at an angle θ to a magnetic field B . State the SI units of all quantities.

ANSWER

$$F = qvB \sin \theta; F \text{ (N)}, q \text{ (C)}, v \text{ (m s}^{-1}\text{)}, B \text{ (T)}$$

RULES YOU NEED

- $F = q v B \sin \theta$
- $v \perp B$: $F = qvB$ (maximum)
- $v \parallel B$: $F = 0$

WORKING — STEP BY STEP

1 The expression

$$F = q v B \sin \theta$$

2 Units

F in newtons (N); q in coulombs (C); v in m s^{-1} ; B in tesla (T); θ in degrees or radians.

3 Sense check with the special cases

$\theta = 90^\circ$ (v perpendicular to B): $\sin \theta = 1$, so $F = qvB$, the biggest possible force.

$\theta = 0$ (v along B): $\sin \theta = 0$, so there is no force at all.

Note on the lecture material: The lecture notes give the Lorentz force for $v \perp B$ only ($F = BQv$) and note that a velocity component along B produces no force (helical motion). The general form with $\sin \theta$ goes slightly beyond this.

Marks	Awarded for
1	$F = qvB \sin \theta$.
1	All SI units.

5(A)(b)**[2 marks]**

A positive charge $q = +10 \mu\text{C}$ moves at 800 m/s perpendicular to a field of 1.2 T . What is the magnitude of the Lorentz force on q ?

ANSWER

$$\mathbf{F = 9.6 \times 10^{-3} \text{ N}}$$

RULES YOU NEED

- $\mathbf{v \perp B: F = q v B}$
- $\mathbf{1 \mu\text{C} = 10^{-6} \text{ C}}$

WORKING — STEP BY STEP**1 Perpendicular, so $\sin \theta = 1$**

$$\mathbf{F = q v B}$$

2 Convert the charge

$$\mathbf{q = 10 \mu\text{C} = 10 \times 10^{-6} \text{ C} = 1 \times 10^{-5} \text{ C}}$$

3 Substitute

$$\mathbf{F = 1 \times 10^{-5} \times 800 \times 1.2 = 1 \times 10^{-5} \times 960 = 9.6 \times 10^{-3} \text{ N}}$$

Marks	Awarded for
1	$\mathbf{F = qvB}$ with μC converted.
1	$\mathbf{9.6 \times 10^{-3} \text{ N}}$.

5(A)(c)**[2 marks]**

A negative charge q travels in the x -direction. A magnetic field B pointing in the y -direction (into the page) is switched on. What is the direction of the force on q ?

ANSWER

Along $-z$ (down the page)

RULES YOU NEED

- Left Hand Rule: First finger = Field, second finger = Current, thumb = Motion (force)
- A negative charge moving one way = conventional current the opposite way

WORKING — STEP BY STEP**1 Turn the moving charge into a current**

Conventional current is the flow of positive charge. A **negative** charge moving along $+x$ is the same as a current along $-x$ (to the left).

2 Apply the Left Hand Rule

First finger into the page (B), second finger to the left (current). The thumb points **down the page**: along $-z$.

3 Cross-check

A positive charge moving along $+x$ would be pushed up ($+z$). Changing the sign of the charge reverses the force, giving $-z$. Both methods agree.

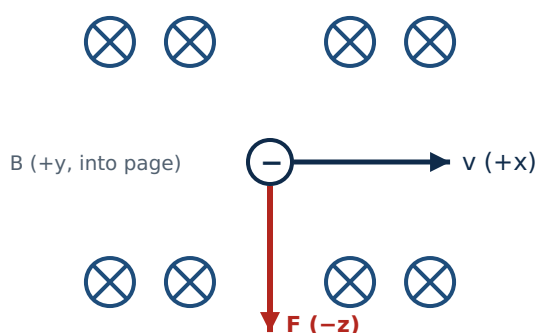


Fig. 2 — A negative charge moving along $+x$ through B into the page is pushed down, along $-z$.

Marks	Awarded for
1	Uses current direction opposite to the electron's motion (or reverses the positive-charge result).
1	$-z$.

5(A)(d)(i)**[2 marks]**

Two parallel wires, each 4 m long, carry the same current and are 0.1 m apart in a vacuum. The force on each is 2×10^{-4} N. What is the current I in each wire?

ANSWER

$$I = 5 \text{ A}$$

RULES YOU NEED

- Force per unit length: $F / \ell = \mu_0 I_1 I_2 / (2\pi d)$
- $\mu_0 / 2\pi = 2 \times 10^{-7} \text{ T m A}^{-1}$

WORKING — STEP BY STEP**1 Same current in both, so $I_1 I_2 = I^2$**

$$F / \ell = (\mu_0 / 2\pi) \cdot I^2 / d$$

2 Force per metre

$$F / \ell = 2 \times 10^{-4} \text{ N} \div 4 \text{ m} = 5 \times 10^{-5} \text{ N m}^{-1}$$

3 Rearrange for I^2

$$I^2 = (F/\ell) \times d \div (\mu_0/2\pi)$$

4 Substitute

$$I^2 = 5 \times 10^{-5} \times 0.1 \div 2 \times 10^{-7} = 5 \times 10^{-6} \div 2 \times 10^{-7} = 25$$

5 Square root

$$I = \sqrt{25} = 5 \text{ A}$$

EXAM TECHNIQUE

Remember to divide by the length: the formula gives force *per metre*, and the question gives the total force on a 4 m wire.

Marks	Awarded for
1	Correct formula $F/\ell = \mu_0 I^2 / (2\pi d)$.
1	$I = 5 \text{ A}$.

5(A)(d)(ii)**[2 marks]**

If the current in each wire is tripled, by what factor does the force change?

ANSWER

The force increases by a factor of 9

RULES YOU NEED

- $F \propto I_1 I_2$

WORKING — STEP BY STEP**1 How F depends on the currents**

F contains one factor of current from each wire: $F \propto I_1 I_2 = I^2$.

2 Triple both

$$F_{\text{new}} / F = (3I \times 3I) / (I \times I) = 3^2 = 9$$

3 Check

New force = $9 \times 2 \times 10^{-4} = 1.8 \times 10^{-3} \text{ N}$.

Marks	Awarded for
1	$F \propto I_1 I_2 = I^2$.
1	Factor 9.

Question 5(A) Total: 10 Marks

Question 5 · Option B

Current, resistance and power; a series-parallel circuit — 10 marks

Given: Circuit: $V = 150 \text{ V}$, $R_1 = 12 \Omega$ in series with $R_2 = 6 \Omega$ and $R_3 = 6 \Omega$ in parallel.

5(B)(a)

[3 marks]

Define and give the SI unit of (i) electric current (ii) electrical resistance (iii) electrical power.

ANSWER

$$I = Q/t \text{ (A)}; R = V/I \text{ (}\Omega\text{)}; P = IV \text{ (W)}$$

RULES YOU NEED

- $I = Q / t$
- $R = V / I$
- $P = E / t = IV$

WORKING — STEP BY STEP

1 Current

The rate of flow of charge: $I = Q/t$. Unit: the **ampere, A** ($1 \text{ A} = 1 \text{ C s}^{-1}$).

2 Resistance

How strongly a conductor opposes the flow of charge: $R = V/I$. Unit: the **ohm, Ω** ($1 \Omega = 1 \text{ V A}^{-1}$).

3 Power

The rate at which electrical energy is used or delivered: $P = E/t = IV$. Unit: the **watt, W** ($1 \text{ W} = 1 \text{ J s}^{-1}$).

Marks	Awarded for
1	Current, A.
1	Resistance, Ω .
1	Power, W.

5(B)(b)**[2 marks]**

In $\Delta t = 5 \text{ s}$, $n = 10^{18}$ electrons flow left to right through the area at X. (i) What is the magnitude of the current? (ii) What is the direction of the conventional current?

ANSWER

(i) $I = 0.032 \text{ A}$ (32 mA); (ii) right to left

RULES YOU NEED

- $Q = n e$, with $e = 1.602 \times 10^{-19} \text{ C}$
- $I = Q / \Delta t$
- Conventional current = direction of positive charge flow

WORKING — STEP BY STEP**1 Total charge**

$$Q = n e = 10^{18} \times 1.602 \times 10^{-19} \text{ C} = 0.1602 \text{ C}$$

2 Current

$$I = Q / \Delta t = 0.1602 \text{ C} \div 5 \text{ s} = 0.032 \text{ A} = 32 \text{ mA}$$

3 Direction

Conventional current is defined as the way **positive** charge would flow. Electrons are negative, so the conventional current is opposite to their motion: **right to left**.

Marks	Awarded for
1	$I = 0.032 \text{ A}$.
1	Right to left (opposite to the electron flow).

5(B)(c)**[5 marks]**

(i) Combined resistance of R_2 and R_3 in parallel? (ii) Total resistance? (iii) Current through R_1 ? (iv) Voltage drop across R_1 ? (v) Total power dissipated?

ANSWER

(i) 3 Ω (ii) 15 Ω (iii) 10 A (iv) 120 V (v) 1500 W

RULES YOU NEED

- Parallel: $1/R = 1/R_2 + 1/R_3$
- Series: $R = R_1 + R_{23}$
- $V = I R$
- $P = V I$

WORKING — STEP BY STEP**1 (i) R_2 and R_3 in parallel**

$$1/R_{23} = 1/6 + 1/6 = 2/6 = 1/3 \Rightarrow R_{23} = 3 \Omega$$

Check: two equal resistors in parallel give half of one, and half of 6 Ω is 3 Ω .

2 (ii) Add R_1 in series

$$R_{\text{total}} = R_1 + R_{23} = 12 + 3 = 15 \Omega$$

3 (iii) Current from the battery

R_1 is in series with everything else, so the whole current passes through it:

$$I = V / R_{\text{total}} = 150 \text{ V} \div 15 \Omega = 10 \text{ A}$$

4 (iv) Voltage across R_1

$$V_1 = I R_1 = 10 \text{ A} \times 12 \Omega = 120 \text{ V}$$

That leaves $150 - 120 = 30 \text{ V}$ across the parallel pair (check: $10 \text{ A} \times 3 \Omega = 30 \text{ V} \checkmark$).

5 (v) Total power

$$P = V I = 150 \text{ V} \times 10 \text{ A} = 1500 \text{ W}$$

Check: $I^2 R_{\text{total}} = 100 \times 15 = 1500 \text{ W} \checkmark$.

Note on the lecture material: The lecture notes don't cover combining resistors in series or parallel ($R = R_1 + R_2$; $1/R = 1/R_1 + 1/R_2$). These formulas come from outside the lecture material; the $V = IR$ and $P = IV$ steps are from the notes.

Marks	Awarded for
1	(i) 3 Ω .
1	(ii) 15 Ω .
1	(iii) 10 A.
1	(iv) 120 V.
1	(v) 1500 W (own-figure accepted throughout).

Question 5(B) Total: 10 Marks

Question 6 · Option A

Specular and diffuse reflection, refractive index, TIR, the eye as a lens — 10 marks

6(A)(a)

[2 marks]

Explain the difference between specular and diffuse reflection. Draw a simple diagram to illustrate this difference.

ANSWER

Specular: smooth surface, parallel rays reflected together.
Diffuse: rough surface, rays scattered in many directions.

RULES YOU NEED

- Law of reflection: $\theta_i = \theta_r$, measured from the normal

WORKING — STEP BY STEP

1 Specular reflection

Happens at a **smooth** surface, such as a mirror. The normal points the same way everywhere, so parallel incoming rays all leave in the same direction. A clear image forms.

2 Diffuse reflection

Happens at a **rough** surface, such as paper. The normal points in different directions at different spots, so parallel rays are scattered in many directions. No image forms.

3 The key point

Each individual ray still obeys $\theta_i = \theta_r$ in both cases. The difference is only the surface.

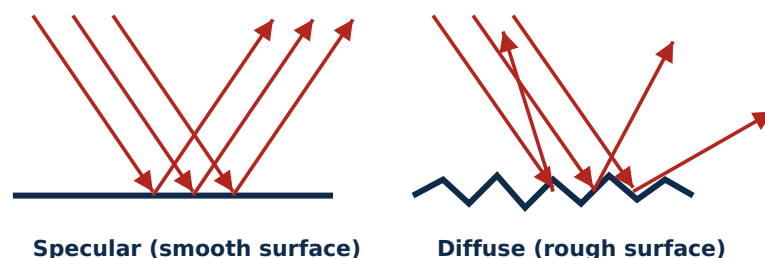


Fig. 3 — Parallel rays stay parallel after reflecting off a smooth surface, but scatter in many directions off a rough one.

Note on the lecture material: Specular vs diffuse reflection is **not covered in the PY1010 lecture notes**, which give only the laws of reflection. This answer is included for completeness.

Marks	Awarded for
1	Correct distinction (smooth vs rough, one direction vs scattered).
1	Diagram.

6(A)(b)(i)**[2 marks]**

A ray exits glass into air. It is incident on the glass-air interface at 20° to the normal and exits at 55.17° . What is the refractive index of the material?

ANSWER

$$n = 2.40$$

RULES YOU NEED

- Snell's law: $n_1 \sin \theta_1 = n_2 \sin \theta_2$
- $n_{\text{air}} = 1$

WORKING — STEP BY STEP**1 Label the two sides**

Side 1 is the glass: $n_1 = n$ (unknown), $\theta_1 = 20^\circ$. Side 2 is air: $n_2 = 1$, $\theta_2 = 55.17^\circ$.

2 Write Snell's law

$$n \sin 20^\circ = 1 \times \sin 55.17^\circ$$

3 Find the sines

$$\sin 20^\circ = 0.3420; \sin 55.17^\circ = 0.8208$$

4 Solve

$$n = 0.8208 \div 0.3420 = 2.40$$

5 Check

The ray bends **away** from the normal as it enters air ($55^\circ > 20^\circ$), which is right for going from a denser to a less dense medium, so $n > 1$ makes sense.

EXAM TECHNIQUE

Make sure your calculator is in **degrees** mode before taking sines.

Marks	Awarded for
1	Snell's law set up correctly.
1	$n = 2.40$.

6(A)(b)(ii)**[2 marks]**

At what angle of incidence would the light be totally reflected back into the glass at the interface? What is this phenomenon called?

ANSWER

Above $\theta_{\text{crit}} \approx 24.6^\circ$: total internal reflection

RULES YOU NEED

- At the critical angle the refracted ray runs along the surface: $\theta_2 = 90^\circ$
- $\sin \theta_{\text{crit}} = n_{\text{air}} / n_{\text{glass}}$

WORKING — STEP BY STEP**1 Set $\theta_2 = 90^\circ$ in Snell's law**

$$n \sin \theta_{\text{crit}} = 1 \times \sin 90^\circ = 1$$

2 Rearrange

$$\sin \theta_{\text{crit}} = 1 / 2.40 = 0.4167$$

3 Inverse sine

$$\theta_{\text{crit}} = \sin^{-1}(0.4167) = 24.6^\circ$$

4 Name it

For any angle of incidence greater than 24.6° , no light gets out: it is all reflected back into the glass. This is **total internal reflection**.

Marks	Awarded for
1	$\theta_{\text{crit}} \approx 24.6^\circ$ (own-figure accepted).
1	Total internal reflection.

6(A)(c)(i)**[2 marks]**

Treat the eye as a single thin lens 2.5 cm from the retina. If the eye is focused on an object at infinity, what should its focal length be to focus the image on the retina?

ANSWER

$$f = 2.5 \text{ cm}$$

RULES YOU NEED

- Thin lens equation: $1/u + 1/v = 1/f$
- Object at infinity: $1/u \rightarrow 0$

WORKING — STEP BY STEP**1 Identify u and v**

The object is at infinity, so $u \rightarrow \infty$ and $1/u \rightarrow 0$. The image must land on the retina, so $v = 2.5 \text{ cm}$.

2 Substitute

$$1/f = 0 + 1/2.5$$

3 Solve

$$f = 2.5 \text{ cm}$$

4 Makes sense?

Yes: light from very far away focuses at the focal point, so the focal point must sit on the retina.

Marks	Awarded for
1	$1/u \rightarrow 0$ for an object at infinity.
1	$f = 2.5 \text{ cm}$.

6(A)(c)(ii)**[2 marks]**

If the object is brought to 25 cm from the lens (the near point), what does the new focal length need to be to keep the object in focus?

ANSWER

$f \approx 2.3$ cm (2.27 cm): the lens must become more strongly converging

RULES YOU NEED

- $1/u + 1/v = 1/f$
- The retina doesn't move: v stays 2.5 cm

WORKING — STEP BY STEP**1 New values**

$u = 25$ cm (the object), $v = 2.5$ cm (the retina).

2 Substitute

$$1/f = 1/25 + 1/2.5 = 0.04 + 0.40 = 0.44 \text{ cm}^{-1}$$

3 Invert

$$f = 1 \div 0.44 = 2.27 \text{ cm}$$

4 Interpret

The focal length is shorter than 2.5 cm, so the lens must become **more strongly converging** (the eye muscles make it fatter) to focus on a near object.

EXAM TECHNIQUE

Keep every length in the same unit (cm here). You don't need to convert to metres as long as you're consistent.

Marks	Awarded for
1	Thin lens equation with $u = 25$ cm, $v = 2.5$ cm.
1	$f \approx 2.27$ cm.

Question 6(A) Total: 10 Marks

Question 6 · Option B

Types of wave, $v = f\lambda$, interference of water waves from two stones — 10 marks

Given: Stones land 4 m in front of you and 4 m out but 3 m to the right; wave period 0.8 s; crest spacing 2 m.

6(B)(a)

[2 marks]

Explain the difference between transverse and longitudinal waves, with an example of each.

ANSWER

Transverse $\perp$ to propagation (light); longitudinal $\parallel$ to propagation (sound)

RULES YOU NEED

- Compare the direction the medium moves with the direction the wave travels

WORKING — STEP BY STEP

1 Transverse

The medium moves **perpendicular** (at right angles) to the direction the wave travels. Example: light, or ripples on water.

2 Longitudinal

The medium vibrates **parallel** to the direction the wave travels, making compressions and rarefactions. Example: sound in air.

Marks	Awarded for
1	Correct distinction.
1	One example of each.

6(B)(b)**[2 marks]**

Orchestras tune to A at 440 Hz. With a speed of sound of 340 m/s, what is the wavelength of this note in air?

ANSWER

$$\lambda \approx 0.77 \text{ m}$$

RULES YOU NEED

$$\bullet v = f \lambda \Rightarrow \lambda = v / f$$

WORKING — STEP BY STEP**1 Rearrange $v = f\lambda$**

$$\lambda = v / f$$

2 Substitute

$$\lambda = 340 \text{ m s}^{-1} \div 440 \text{ Hz} = 0.773 \text{ m}$$

3 Check the units

$$(\text{m s}^{-1}) \div (\text{s}^{-1}) = \text{m} \checkmark.$$

Marks	Awarded for
1	$\lambda = v/f.$
1	0.77 m.

6(B)(c)(i)-(ii)**[3 marks]**

(i) Calculate the frequency of the wave in SI units. (ii) Calculate the velocity of the wave.

ANSWER

(i) $f = 1.25 \text{ Hz}$ (ii) $v = 2.5 \text{ m s}^{-1}$

RULES YOU NEED

- $f = 1 / T$
- Distance between successive crests = wavelength λ
- $v = f \lambda$

WORKING — STEP BY STEP**1 (i) Frequency from the period**

$$f = 1 / T = 1 \div 0.8 \text{ s} = 1.25 \text{ Hz}$$

2 (ii) Spot the wavelength

The distance between one crest and the next is exactly one wavelength: $\lambda = 2 \text{ m}$.

3 Multiply

$$v = f \lambda = 1.25 \text{ Hz} \times 2 \text{ m} = 2.5 \text{ m s}^{-1}$$

4 Check

Another way: one wavelength (2 m) passes every period (0.8 s), so $v = 2 \div 0.8 = 2.5 \text{ m s}^{-1} \checkmark$.

Marks	Awarded for
1	$f = 1/P = 1.25 \text{ Hz}$.
1	$\lambda = 2 \text{ m}$ identified.
1	$v = 2.5 \text{ m s}^{-1}$.

6(B)(c)(iii)-(iv)**[3 marks]**

(iii) Calculate the direct distance d from where the second stone lands to where you are standing. (iv) When the first waves from each stone meet in front of you, will a large wave or no wave be produced? Explain.

ANSWER

(iii) $d = 5 \text{ m}$ (iv) Path difference = $\lambda/2$, so destructive interference: no wave

RULES YOU NEED

- Pythagoras: $d = \sqrt{a^2 + b^2}$
- Path difference of a whole number of $\lambda \Rightarrow$ in phase (constructive)
- Path difference of half a $\lambda \Rightarrow$ out of phase (destructive)

WORKING — STEP BY STEP**1 (iii) Draw the triangle**

The second stone is 4 m out and 3 m to the side. These two sides are at right angles, so the direct distance is the hypotenuse.

2 Pythagoras

$$d = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5 \text{ m}$$

3 (iv) Path difference

The first stone is 4 m away and the second is 5 m away. Both land at the same time, so their waves start in phase.

$$\Delta = 5 \text{ m} - 4 \text{ m} = 1 \text{ m}$$

4 Compare with the wavelength

$$\Delta / \lambda = 1 \text{ m} \div 2 \text{ m} = \frac{1}{2}$$

The second wave has travelled exactly half a wavelength further.

5 Conclusion

Half a wavelength behind means the waves arrive **exactly out of phase**: a crest from one meets a trough from the other. They cancel (destructive interference), so **no wave** is produced where you stand.

Note on the lecture material: The notes show that waves exactly out of phase cancel, but don't state the path-difference condition $(n + \frac{1}{2})\lambda$ explicitly. The reasoning here combines the lectures' superposition pages with λ as the crest spacing.

Marks	Awarded for
1	$d = 5 \text{ m}$.
1	Path difference $= 1 \text{ m} = \lambda/2$.
1	Destructive: no wave.

Question 6(B) Total: 10 Marks